

Eric Li

778-929-1402 | eric.qli9@gmail.com | [LinkedIn](#) | [GitHub](#)

Education

University of Toronto

Sep 2021 – Jun 2026

BSc, Computer Science (Specialist), Statistics (Major)

Toronto, ON

- Relevant Coursework: Operating Systems, Databases, AI, NLP

Experience

Data Engineer

May 2024 – April 2025

Royal Bank of Canada

Toronto, ON

- **Change Data Capture (CDC)**: Designed and implemented a **Python (Pandas)** CDC pipeline to clean and transform large datasets, generate surrogate keys, and keep records in near-real-time sync across multiple sources. Separated new, updated, and closed units for downstream systems.
- **Transit Hierarchy Cleaning**: Re-engineered the transit hierarchy used for risk assessment by matching transits across legal entity, location, and financial metrics with **Python/Pandas**. Built a **React**-based validation app and backend services to manage hierarchy changes and improve data quality.
- **NLP for Incident Reports**: Built an end-to-end **NLP pipeline** in **spaCy** to convert free-text incident reports into structured fields. Labeled a historical dataset and trained a **supervised text-classification model** to auto-tag event type, business line, and root cause, improving searchability and reuse of past risk events.

Software Engineer

May 2023 – August 2023

Deutsche Telekom

Beijing, China

- **Office Automation Website Development**: Built an office automation web application using the Cumanada workflow engine, implementing Java backend services, CRUD operations, and workflow APIs.

Projects

Mini Operating System Kernel

Feb 2026 – Present

C, CMake, Unicorn Engine

- Implemented a Unix-inspired process management kernel in **C**, including **fork**, **exec**, **wait**, system call handling, process control blocks (PCBs), and cooperative context switching on an emulated x86-64 CPU.

AI Job Application Platform & Personal Portfolio

May 2026 – Present

Next.js, TypeScript, Supabase, PostgreSQL, OpenAI API, Vercel

- Built a full-stack AI-powered job application platform and personal portfolio using **Next.js**, **TypeScript**, **Supabase**, and **PostgreSQL**.
- Implemented a job tracker with CRUD operations and integrated **OpenAI** APIs for job description analysis, tailored cover letter generation, and resume recommendations.
- Designed a responsive UI and deployed the application on **Vercel** for scalable access and application management.

Scriptorium Code Template Manager

Jan 2026 – April 2026

Next.js, Node.js, TypeScript, JavaScript, Prisma, PostgreSQL

- Built a multi-language code template platform using **Next.js** and **Node.js**.
- Developed backend APIs with **Prisma**, **PostgreSQL**, JWT authentication, and Docker.
- Integrated **OpenAI** APIs for AI-assisted debugging, error explanation, and code correction.

Technical Skills

- **Languages**: Python, Java, C/C++, JavaScript/TypeScript, SQL
- **Frameworks**: React, Node.js, Express, Django/Flask
- **Tools**: Git, Docker, Linux, VS Code, IntelliJ